DEPARTMENT OF ELECTRONICS & COMM. ENGG
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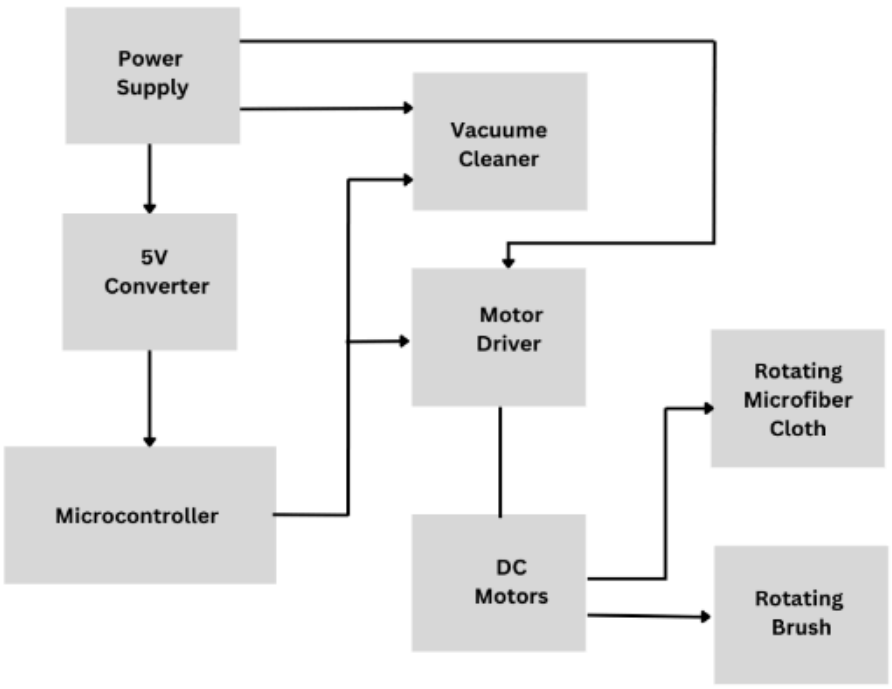
MINI PROJECT EXHIBITION 2024
DRY SOLAR PANEL CLEANING SYSTEM

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Preamble

In pursuit of advancing solar panel maintenance, this project introduces an automated dry cleaning system aimed at boosting efficiency while preserving panel integrity. Through a detailed examination of current cleaning techniques, we are developing an innovative, water-free solution. The system will incorporate a vacuum cleaner, brush, and microfiber cloth to minimize human intervention and conserve water. Comprehensive testing across different solar panel surfaces will ensure the prototype's effectiveness and guide the refinement of the final design.

Block Diagram



Brief Description

The proposed solar panel cleaning system aims to streamline maintenance by integrating advanced technologies to enhance efficiency while safeguarding photovoltaic surfaces. It combines rotating brushes, vacuum cleaners, and microfiber cloths, driven by geared DC motors, to achieve thorough cleaning without causing damage. The rotating brush dislodges debris, the vacuum cleaner removes loose particles, and the microfiber cloth provides a final wipe for optimal cleanliness. This automated system reduces water usage, minimizes manual labor, and lowers operational costs while maintaining consistent cleaning performance. Feasibility studies and field visits validate its practicality and effectiveness. The final design will be optimized based on testing outcomes, aiming to improve energy yield and extend the lifespan of solar installations, thereby supporting broader adoption of solar technology

Target achieved

- **Enhanced Cleaning Efficiency**: The cleaning system uses rotating brushes, vacuum suction, and microfiber cloths to effectively remove dust and debris from solar panels, significantly improving cleanliness and energy yield.
- **Technical Feasibility:** Confirmed the cleaning mechanism's feasibility, effectively removing dust and debris from various solar panel surfaces and configurations.
- **Design and Implementation:** Successfully developed a solar panel cleaning system prototype using a rotating brush, vacuum cleaner, and microfiber cloth.
- **Energy Efficiency:** The system's cleaning process enhances solar panel efficiency, resulting in increased energy output and prolonged panel lifespan.

Future Scope

- **Complete Autonomy:** Future plans involve making the cleaning system fully autonomous, minimizing the need for human intervention and ensuring consistent maintenance.
- **IoT Integration:** We aim to integrate IoT technology for remote monitoring and control, optimizing cleaning schedules based on real-time data and environmental conditions.

